

Nathan Au

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Education

Concordia University

Bachelor of Engineering – Software Engineering, Co-op

Montreal, QC

Sep. 2024 – May 2028 (expected)

Relevant Skills

Technical Languages: Python, TypeScript, JavaScript, SQL, Dart, HTML, CSS, C++, Java, Bash

Frameworks & Libraries: FastAPI, Flutter, React, Next.js, PyTorch, Hugging Face Transformers, GeoPandas, Shapely, GDAL, rasterio, NetworkX, MapLibre GL, Leaflet, PMTiles, pyproj, PuLP, HiGHS, Node.js, Tailwind CSS, Pandas, Matplotlib, NumPy, Pytest, Vitest, Playwright, OpenCV, Ultralytics YOLO, Google ADK, Pydantic, LangChain, LiveKit, Zod, Stripe, Twilio

Practices: OOP, REST APIs, Agile Scrum, Git Version Control, Unit & E2E Testing, ETL Data Pipelines, LLM Integration, Agentic Systems, Prompt Engineering, CI/CD, Database Design, System Design, Docker Containerization, Integer Linear Programming, Semantic Segmentation, Mixed-Precision Training (AMP), Model Evaluation & Cross-Validation, WebAssembly, Web Workers, Concurrency & Parallelism, Geospatial Data Processing, CRS & Projection Handling, Parquet, HTTP Protocol Semantics, AWS (EC2, S3, IAM), Linux

Spoken Languages: English (Native), French (Professional – CECR DELF B2)

Experience

Software Engineering Intern

Royal Bank of Canada

Sep. 2026 – Apr. 2027

Montreal, QC

- Incoming Fall 2026

Software Engineering Intern

Dendrometric

May 2026 – Aug. 2026

Montreal, QC

- Built a geospatial analytics engine in Python and GeoPandas evaluating 286 government timber sales across 23,040 optimizer runs (30 sampling origins $\times$ 4 drone ranges), pairing a set-cover ILP (PuLP/CBC) with a road-network TSP to model a 90% reduction in required field stops (28,625 $\rightarrow$ 2,839) and 13,257 km of ground-crew walking against a deliberately conservative manual baseline.
- Trained a SegFormer-B2 semantic segmentation model (PyTorch, Hugging Face Transformers) on a self-built 292,968-tile dataset from 236 aerial orthophotos, reaching 0.856 mIoU and 0.943 Cohen's κ on a 29,298-tile held-out set at 17.9 ms/tile under a 40:1 class imbalance.
- Ported two Python geospatial pipelines to run entirely client-side in TypeScript and Next.js, swapping GEOS for closed-form analytic geometry and a native ILP solver for HiGHS compiled to WebAssembly, removing any need for backend hosting and verifying sub-centimetre parity with GDAL/pyproj across a 2,594 km road network.
- Built a self-hosted forestry web GIS (Next.js, MapLibre GL) serving 70 data overlays and 115 runtime map layers from 17 PMTiles archives with HTTP Range-aware streaming (206/416, ETag, immutable caching), and engineered the GDAL pipeline converting 1 m lidar DTM, government vectors, and drone orthophotos into web tiles.
- Engineered a fault-tolerant, rate-limited crawler that acquired a 3,777-record government archive of 12,525 documents and 59 GB in a single 5-hour run with zero index failures and 99.97% download success, using HEAD-sizing and atomic staged renames to prevent truncated files from appearing complete.

Software Engineering Intern

Biotics Labs

Jan. 2026 – Apr. 2026

Montreal, QC

- Scaled a conversational Twilio SMS user onboarding pipeline that collects 30+ data points with async message processing for metrics calculation, context generation, and nutrition plan creation to eliminate production timeouts.
- Shipped end-to-end full-stack features from frontend UI (Next.js, shaden/ui) to backend API design (FastAPI), database schema and authentication (Supabase), payment processing (Stripe), and production CI/CD (Railway).
- Architected a LiveKit-based agentic voice system with tool-calling LLMs, custom SessionManagers for persistent multi-turn conversations, and database-integrated tools to autonomously recommend and collect meal orders.
- Deployed production LLM infrastructure using Gemini and LangChain with Langfuse prompt management and context aggregation from user profiles and conversation history to enable personalized interactions at scale.

Software Developer

Google Developer Group Montreal

Nov. 2024 – Apr. 2025

Montreal, QC

- Developed a cross-platform Flutter mobile application that helps university students in Montreal connect with peers, discover local study spots, and stay up-to-date on upcoming campus events all on one platform.

- Built a data-driven recommendation engine using user profiles managed in Firebase Realtime Database and Gemini LLM integration via Vertex AI to generate personalized connection suggestions based on shared interests and goals.
- Integrated the Concordia University Open Data API to retrieve 60+ upcoming campus events in real-time and implemented responsive geolocation-based features using Flutter Map for interactive study spot discovery.
- Delivered technical presentations at Flutter Montreal and Concordia University to audiences of 100+ attendees and produced the official launch video to showcase system architecture and data flow.

Projects

Reel Digest | Python, Telegram Bot API, Ollama (Granite 4), SQLite, yt-dlp, OpenCV | [Source Code](#) **Dec. 2025**
Messaging-based productivity tool that saves users' time by turning short-form content into concise text summaries.

- Designed a multi-modal media processing pipeline with Google STT for audio transcription and OpenCV computer vision/Tesseract OCR optical character recognition for on-screen text extraction to enable contextual LLM analysis.
- Configured a SQLite caching layer to eliminate redundant inference, persist user history, and successfully serve live summaries for 300+ unique production requests.

TeeFour AI | Python, FastAPI, Ollama (Gemma 3), SQLAlchemy, Tesseract OCR, Pytest | [Source Code](#) **Oct. 2025**
Automated tax document processing system that eliminates manual data entry for tax accounting workflows.

- Engineered a two-layer document classification algorithm using PyMuPDF/Tesseract OCR to categorize known document types (T4s, IDs, receipts) via filename analysis and content-keyword matching logic.
- Streamlined tax form data field extraction by integrating a locally hosted LLM to transform unstructured OCR text inputs into validated, machine-readable JSON outputs for downstream use in tax filing software.

More projects: nathan-au.github.io